

JPMORGAN CHASE CODE FOR GOOD 2025

BARN

Benefactor for Agriculture Resources and Need

Smart Agriculture & Rural NGO Management Platform



The Gap Holding Rural Agriculture Back

Small and marginal farmers and rural NGOs rely on paper-based workflows, fragmented communication, and disconnected systems — limiting operational efficiency and making it nearly impossible to scale agricultural support programs.



Manual Record Keeping

No digital audit trails or searchable data



Paper-Based Operations

Slow, error-prone, and non-scalable



Poor Communication

Fragmented channels with no unified hub



Limited Analytics

No data-driven decision making

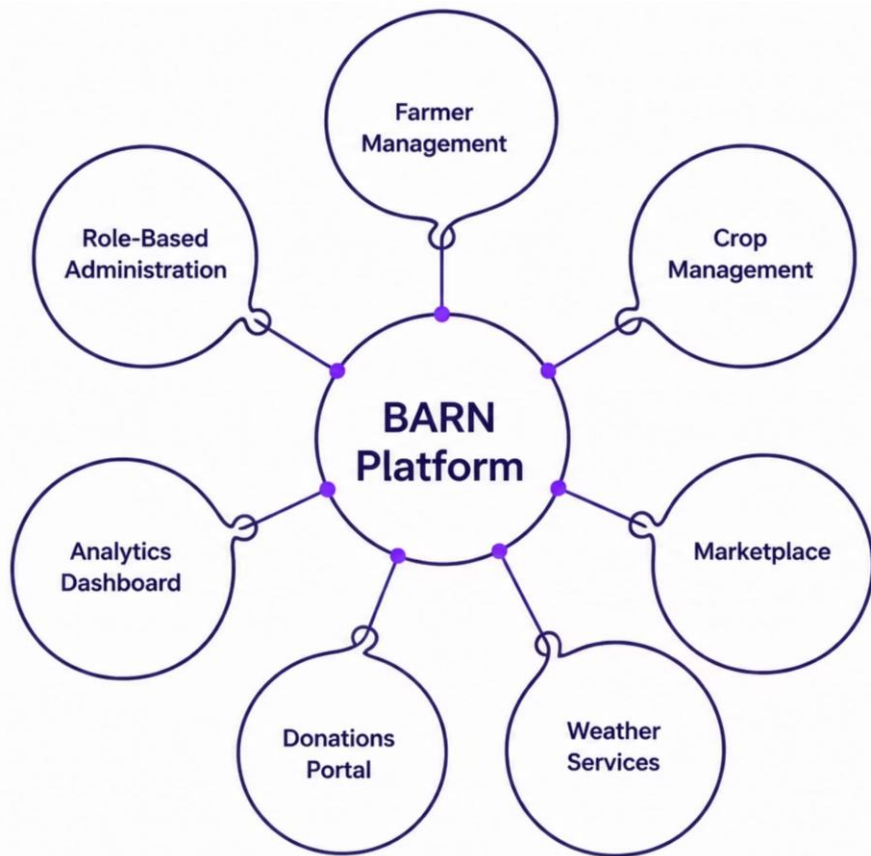


No Unified Platform

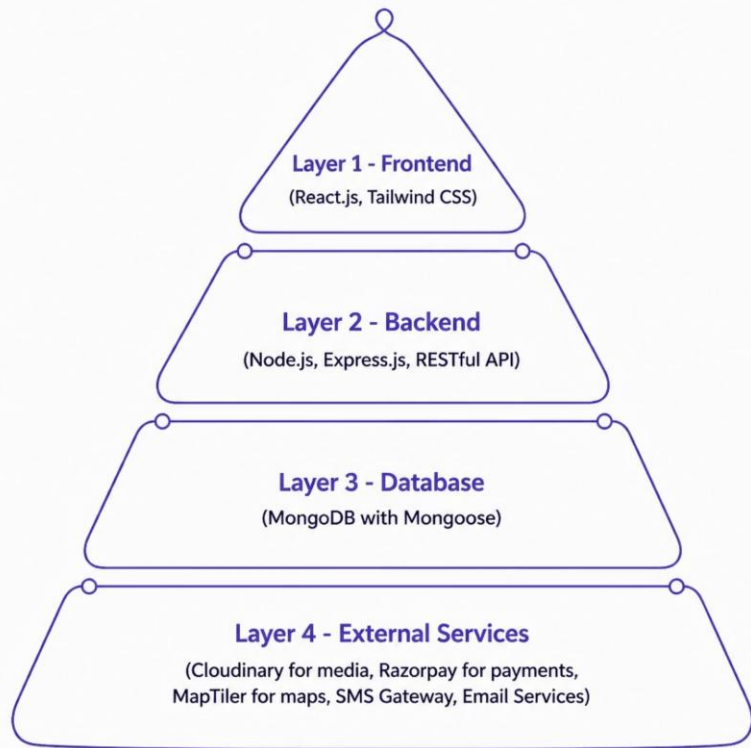
Siloed tools create operational blind spots

One Platform. Every Need Covered.

BARN unifies farmer management, crop tracking, marketplace, weather intelligence, donations, and analytics into a single, role-based SaaS platform — purpose-built for rural NGOs and agricultural stakeholders.



Enterprise-Grade Architecture



Designed for Scale

A clean separation of concerns ensures each layer can evolve independently. The architecture supports high availability, horizontal scaling, and seamless third-party integrations — from payments to geospatial mapping.

Frontend

React.js, Tailwind CSS

Backend

Node.js, Express.js

Database

MongoDB, Mongoose

Cloud Services

Cloudinary, Razorpay, MapTiler

Everything an NGO Needs, In One Place



Farmer Management

Profiles, production tracking, and Excel import/export



Marketplace

Direct agricultural product listings and transactions



Weather Integration

Real-time forecasts and climate alerts for farmers



Analytics Dashboard

Operational insights with Chart.js and Recharts



Crop Management

Seasonal tracking and yield monitoring



Donation Portal

Secure digital giving with Razorpay integration



Real-Time Chat

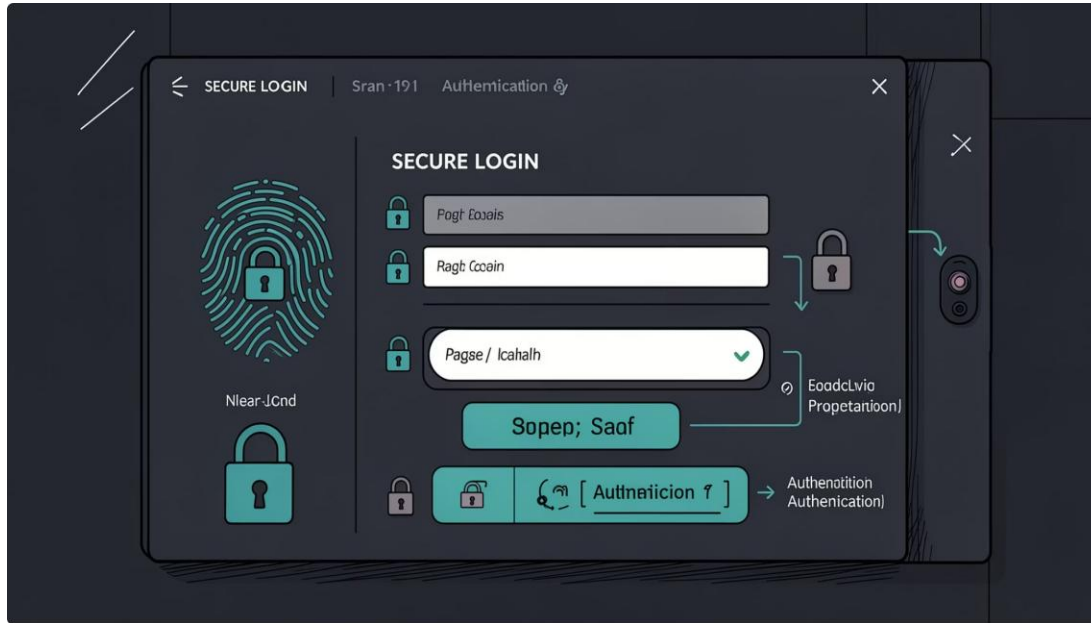
Instant communication between NGOs and farmers



Interactive Maps

MapTiler-powered geospatial visualization

Secure Access for Every Role



→ Email Login

Secure password-based authentication with recovery flows

→ SMS OTP Login

Two-factor verification for high-security operations

→ Google OAuth

One-click social sign-in for seamless onboarding

→ Role-Based Authorization

Admin approval gates and granular access control per role

Multi-Layered Authentication

BARN implements a robust, multi-factor authentication system ensuring only verified users access sensitive agricultural and financial data.

Built With Modern, Proven Technologies



Frontend

React.js · Tailwind CSS



Backend

Node.js · Express.js



Database

MongoDB · Mongoose



Authentication

Google OAuth · SMS OTP



Cloud & Media

Cloudinary · MapTiler



Payments

Razorpay Integration



Visualization

Chart.js · Recharts

Every technology choice was made for reliability, developer experience, and production readiness — ensuring BARN can scale from pilot to enterprise deployment.

Full Stack Ownership Across the Platform

Authentication System

Google OAuth, SMS OTP, password recovery, and secure session management

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Role-Based Access Control

Admin approval workflows and granular permission layers across all modules

Cloudinary Integration

Media upload, optimization, and delivery for farmer profiles and crop imagery

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Payment & Marketplace

Razorpay integration, donation module, and full agricultural marketplace backend

API & Full Stack Integration

RESTful API development, chat system, weather integration, and frontend enhancements

5

Technology That Changes Lives



From Paper to Platform

BARN replaces fragmented manual processes with a unified digital ecosystem — enabling NGOs to operate at scale and farmers to access markets, weather data, and support services instantly.



Farmer Support

Direct access to markets, weather, and NGO resources



Digital Donations

Transparent, secure giving with full transaction tracking



Operational Analytics

Data-driven decisions for NGO program managers



Scalable Rural Tech

Architecture designed to grow from village to region

BARN

Where Full Stack Engineering Meets Social Impact

BARN demonstrates how modern Full Stack Software Engineering can transform rural NGO operations — through digital automation, cloud services, secure authentication, and scalable web technologies.

Complete Platform

End-to-end full stack solution

Role-Based Architecture

Secure, scalable access control

Cloud Integration

Payments, media, maps & SMS

NGO Transformation

Digital operations at scale

Developed during JPMorgan Chase Code for Good 2025 — Thank you. Questions welcome.